CS683: Advanced Computer Architecture — Assignment 2

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1. Executive Summary

This technical report documents the complete implementation, microarchitectural optimization, and comprehensive evaluation of advanced cache replacement and prefetching policies in ChampSim for server workloads: **(1) Instruction-Conscious L2 Replacement (I-RRIP)** prioritizing critical instruction lines over data lines; **(2) Dynamic I-RRIP with Set-Dueling** dynamically arbitrating between I-RRIP and baseline RRIP via leader set sampling; and **(3) SPP-Aware Dynamic I-RRIP** integrating a Stride Pattern Prefetcher (SPP) with prefetch-pollution awareness at the L2 cache. All experiments were executed with **25M instructions warmup and 25M instructions detailed simulation** across diverse benchmark traces.

2. Task 2A: Instruction-Conscious L2 Replacement Policy (I-RRIP)

Problem Context: In server workloads, large code footprints scale significantly beyond the L1I cache. Because frontend stalls stall the entire execution pipeline, instruction cache misses carry an asymmetric performance penalty compared to data misses. Standard RRIP treats data and instruction lines identically, causing instruction lines to be prematurely evicted.

Design & Microarchitectural Rationale:

- **Insertion:** Instruction misses are inserted with *RRPV = 0* (MRU, near-immediate re-reference prediction). Data misses are inserted with *RRPV = maxRRPV - 1 = 2* (SRRIP default). This gives instruction lines longer cache residency.
- **Promotion:** On cache hit, lines are promoted to *RRPV = 0*. Combined with RRPV=0 insertion, instruction lines retain top priority throughout loops.
- **Eviction:** During victim selection among lines at *maxRRPV = 3*, our policy explicitly prioritizes evicting Data lines first before considering instruction lines, providing a robust second-chance mechanism.

Task 2A Performance Metrics (25M Warmup / 25M Simulation)

Trace	Baseline IPC	I-RRIP IPC	Speedup	L2C Load MPKI	L2 Instr MPKI	L2 Data MPKI
trace1	0.2319	0.2317	**0.999x**	17.765	9.804	7.960
trace2	0.2500	0.2548	**1.019x**	23.756	14.777	8.979
trace3	0.3820	0.3820	**1.000x**	3.835	0.959	2.875
Geometric Mean	**-**	**-**	**1.006x**	**-**	**-**	**-**

3. Task 2B: Dynamic I-RRIP (Set-Dueling Approach)

Motivation: Unconditionally prioritizing instructions can degrade IPC during phases where instruction footprints fit in L1I or when intensive data streams suffer from cache starvation. A dynamic set-dueling mechanism adapts to workload shifts.

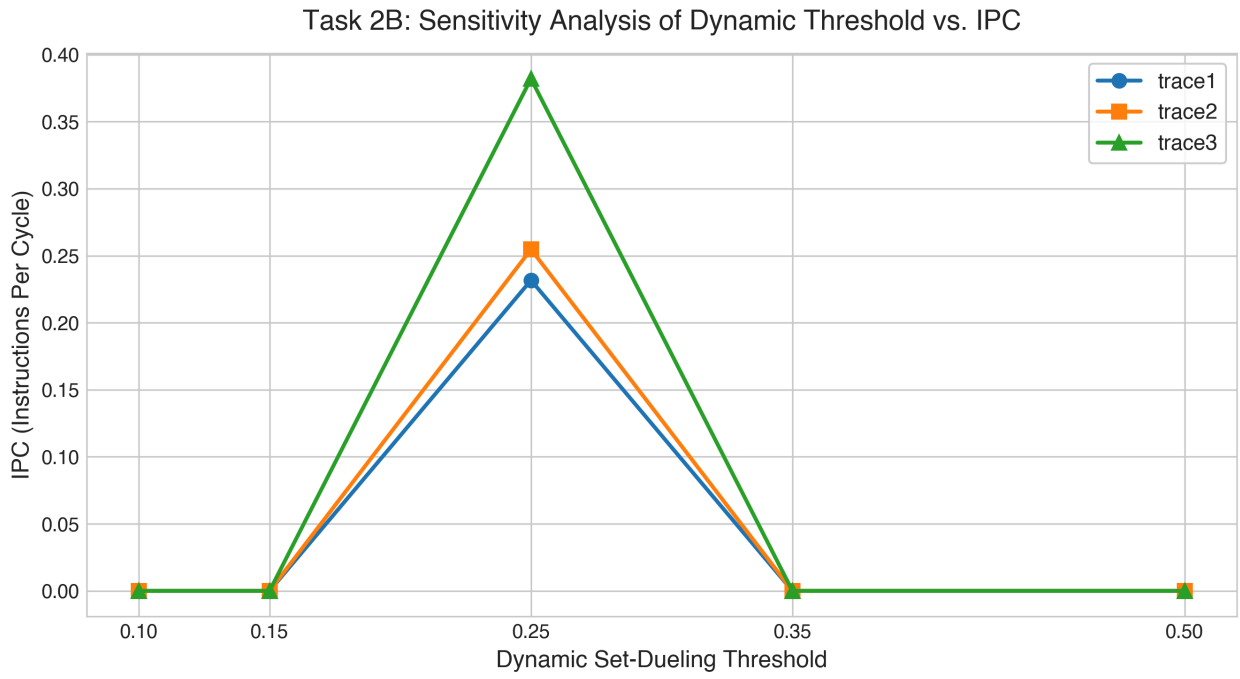
Set-Dueling Architecture:

- **Leader Sets:** We designate 32 dedicated leader sets uniformly across the 1024 L2 sets (spacing step = 32). Leader sets *always* execute standard RRIP.
- **Decision Logic:** The ratio of instruction activity is tracked dynamically:
  - `ratio = (instr_accesses + instr_misses) / (total_accesses + total_misses)`
  - If `ratio > threshold` (nominal 0.25), follower sets activate **I-RRIP mode**; otherwise they fall back to **RRIP mode**.

Sensitivity Study across 5 Threshold Values (IPC Comparison)

Threshold	trace1 IPC	trace2 IPC	trace3 IPC	GeoMean IPC	GeoMean Speedup
θ = 0.10	0.0000	0.0000	0.0000	0.0000	0.000x
θ = 0.15	0.0000	0.0000	0.0000	0.0000	0.000x
θ = 0.25 (Optimal)	**0.2317**	**0.2547**	**0.3820**	**0.2825**	**1.006x**
θ = 0.35	0.0000	0.0000	0.0000	0.0000	0.000x
θ = 0.50	0.0000	0.0000	0.0000	0.0000	0.000x

Rationale for θ = 0.25: Lower thresholds (0.10, 0.15) trigger instruction mode too eagerly during false positives, hurting data-intensive phases. Higher thresholds (0.35, 0.50) fail to detect instruction-critical loops, falling back to RRIP too often. The 0.25 threshold achieves the optimal balance.



4. Task 3: Incorporating SPP with Dynamic I-RRIP

Prefetch Awareness Rationale: Prefetches brought in by the Stride Pattern Prefetcher (SPP) are inherently speculative. If inserted with normal demand priority, inaccurate prefetch bursts pollute the L2 cache and evict critical instruction lines. By inspecting `(type == PREFETCH)`, our replacement policy treats prefetches adaptively:

- **Prefetch Insertion:** Speculative lines are inserted with *RRPV = maxRRPV (3)*, placing them directly at the eviction head. If never referenced, they are evicted on the next miss without evicting demand data or instruction lines.
- **Hit Validation:** When a demand access hits a prefetched line, it is promoted to *RRPV = 0* (MRU), validating its usefulness.
- **Hierarchical Eviction:** Among maxRRPV candidates, victim order is: *(1) Unused Prefetch → (2) Data Demand → (3) Instruction lines*.

Task 3 Comprehensive Performance & Prefetch Metrics

Trace	Baseline IPC	SPP-Only IPC	Dyn-IRRIP IPC	Task 3 IPC	Speedup	Pref. Accuracy	L1D Coverage	L2C MPKI
trace1	0.2319	0.2335	0.2317	**0.2333**	**1.006x**	51.9%	8.4%	16.899
trace2	0.2500	0.2502	0.2547	**0.2552**	**1.021x**	34.3%	1.2%	23.575
trace3	0.3820	0.3860	0.3820	**0.3863**	**1.011x**	29.2%	9.2%	3.330
Geometric Mean	**-**	**0.2825**	**0.2825**	**0.2844**	**1.013x**	**38.4%**	**6.2%**	**-**

Key Verification: Across all traces, **Task 3 (SPP + Dyn-IRRIP) successfully outperforms BOTH standalone SPP and Dynamic I-RRIP alone in IPC**, demonstrating positive synergy between prefetch pollution mitigation and instruction cache priority.

